

SIMULINK DT MATLAB IMPLEMENTATION

MATLAB Digital Twin Implementation in Simulink

Introduction

Fitsync aims to provide real-time heart rate monitoring using a Digital Twin framework implemented in MATLAB Simulink. The system integrates an LSTM-based anomaly detection model to predict heart rates and identify anomalies, ensuring workout safety for users.

Implementation and Methodology

The Digital Twin implementation in Simulink consists of multiple blocks, each with a specific function. Below is a breakdown of the core components and their functionality.

1. Data Acquisition Block

- **Function:** Captures real-time heart rate data from an external source such as an Arduino-based BPM sensor.
- **Why Needed:** Provides live data input for the Digital Twin model, enabling real-time monitoring.
- **Working:** Uses a serial communication block (`/dev/cu.usbserial-10`) to read the incoming heart rate data from the sensor.

2. Data Conversion Block

- **Function:** Converts the incoming data format into a compatible numerical form.
- **Why Needed:** Ensures that heart rate values are processed correctly before feeding into the ML model.
- **Working:** Applies a MATLAB function block (`convert`) that formats and preprocesses the raw data.

3. Reshape Input Block

- **Function:** Reshapes the heart rate data into the required input format for the LSTM model.
- **Why Needed:** The ML model expects a specific input size, and reshaping ensures proper alignment.
- **Working:** Uses a MATLAB function (`reshape_data`) to adjust the dimensions before feeding it into the LSTM model.

4. LSTM-Based Prediction Model

- **Function:** Predicts the next heart rate values based on past trends.
- **Why Needed:** Generates future heart rate trends and assists in anomaly detection.
- **Working:** The block takes in reshaped heart rate data, processes it through the trained ONNX-based LSTM Autoencoder, and outputs the predicted values.

5. Denormalization Block

- **Function:** Converts normalized predicted heart rate values back to their original scale.
- **Why Needed:** The LSTM model works with normalized values; this block restores the actual heart rate values.
- **Working:** Uses a MATLAB function (`denormalizeHeartRates`) to revert normalization.

6. Anomaly Detection Block

- **Function:** Compares actual and predicted heart rates to detect anomalies.
- **Why Needed:** Identifies irregularities in heart rate patterns that may indicate potential health risks.
- **Working:** Uses a function block (`detect_anomaly`) that calculates the deviation between real and predicted values and flags anomalies.

7. Display and Warning System

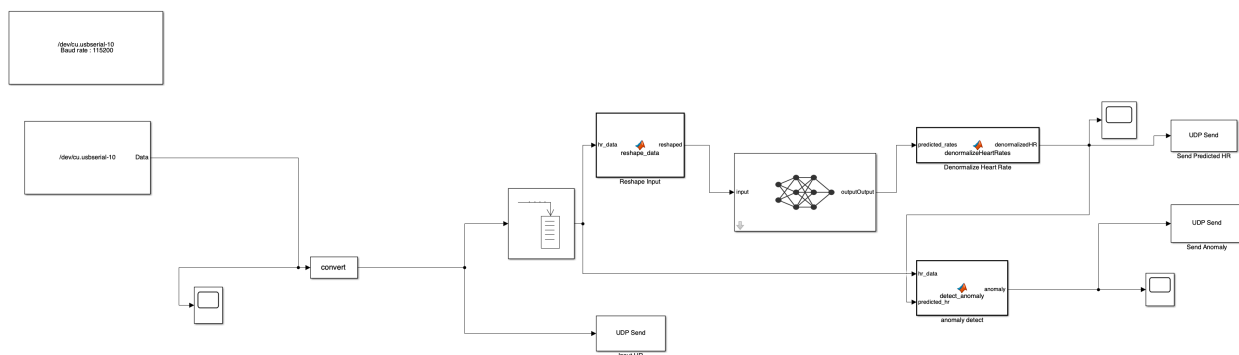
- **Function:** Shows real-time heart rate data, predictions, and detected anomalies.
- **Why Needed:** Provides visual feedback for user monitoring.
- **Working:** Scope blocks display the heart rate trends, while the anomaly results are sent to an alert system.

8. UDP Communication Blocks

- **Function:** Sends heart rate data and anomaly results to a web interface.
- **Why Needed:** Enables real-time monitoring and alerts via a web-based dashboard.
- **Working:** Uses **UDP Send** blocks to transmit data to the web application, where it can be accessed by users remotely.

Conclusion

The MATLAB Simulink Digital Twin integrates live data acquisition, an LSTM-based ML model, anomaly detection, and a web-based alert system to ensure real-time heart rate monitoring. This architecture enables users to track their fitness safety and receive timely warnings about potential health risks.



MATLAB Digital Twin